Haoran Xu

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Education

Zhejiang University 2024 – June 2028 (expected)

Undergraduate, Control Science and Engineering

GPA: **4.22/5.00** | Weighted Average: **87.30/100**

TOEFL iBT: 5.5/6.0 (2026 scale)

Research Experience

Machine Learning Research 2026 – Present

Research on learning processes, multimodal model behavior, and probabilistic prediction.

- Analyzed the learning dynamics of sequential model updates and their effects on later model behavior, combining mathematical reasoning with experiments on test-time adaptation.
- Investigated how vision-language models use visual information, using internal interventions to distinguish available information from its influence on model behavior.
- Studied predictive coherence in probabilistic forecasting and examined whether evaluations of predictive distributions capture consistency as well as accuracy.

Two research manuscripts are currently under peer review.

Public preprint: Haoran Xu. *Same Data, Different Domains: Representation Dependence in Adaptive Data Mixing*. Preprint, 2026. https://doi.org/10.5281/zenodo.23428428

Machine Learning for Bioprocess Soft Sensing 2025 – 2026

Zhejiang University | Supervised by Prof. Xinggao Liu

- Studied machine learning-based soft sensing using industrial multivariate time series from erythromycin fermentation processes. Explored regression models for predicting process variables from production data.
- Compared prediction errors across target variables and examined changes associated with time, production conditions, and data distributions. Analyzed distribution shifts to investigate the reliability and limitations of regression predictions across process settings.

Honors & Awards

Zhejiang University Scholarship, Second Prize, 2024–2025.